Comprehensive Analytical Compendium of the Scientific Career and Biographical History of Wylie Walker Vale Jr.

The evolution of neuroendocrinology from a discipline of theoretical hypotheses to a rigorous field of molecular characterization is inextricably linked to the professional trajectory of Wylie Walker Vale Jr. Over a career spanning nearly five decades, Vale transitioned the study of brain-body communication from the macroscopic observations of physiology to the atomic-level specificity of peptide chemistry and receptor signaling.[1] His work at the Salk Institute for Biological Studies did not merely catalog new hormones; it provided a structural framework for understanding how the central nervous system (CNS) perceives environmental challenge and translates that perception into systemic physiological change.[3]

A) Anchor Scenes

SCENE #1: The Odiferous Labor of the Sheep Hypothalamus Extraction

The pursuit of hypothalamic-releasing hormones in the late 1960s and early 1970s was a feat of industrial-scale biochemistry that required a unique blend of intellectual rigor and physical endurance. While Roger Guillemin provided the visionary leadership, Wylie Vale was often at the center of the operational "elbow grease" required to process biological material at an unprecedented scale.[3] The laboratory at Baylor College of Medicine, and later the Salk Institute, functioned as a processing plant for hundreds of thousands of sheep hypothalami.[3]

Component	Metric/Detail	Source
Raw Material	500,000+ Sheep Hypothalami	3
Environment	Temporary wooden barracks west of the main Salk building	3
Sensory Texture	Overpowering smell of organic solvents and biological extracts	3

Social Consequence	Researchers were avoided in supermarkets due to the lingering stench	[3]
Primary Actors	Wylie Vale, Catherine Rivier, Jean Rivier	[3]

The process involved harvesting the tissue, followed by massive-scale extraction using petroleum ether and aqueous buffers. Catherine Rivier recalls that the work was so all-consuming and the chemical byproducts so pungent that the researchers effectively became social pariahs in La Jolla's public spaces.[3] This period represents the "heroic" era of endocrinology, where the existence of Corticotropin-Releasing Factor (CRF) was a matter of intense debate and the technology to sequence it was still in its infancy.[4] The tension was fueled by a fierce international competition with Andrew Schally's laboratory at Tulane, a rivalry so documented that it became known as the "Nobel Duel".[5] Vale's role was to develop the quantitative and selective bioassays that could detect the minute presence of active peptides within these massive extracts, a task requiring a level of precision that many of his peers struggled to maintain.[4]

SCENE #2: The 1981 April Fools' Deception and the High-Stakes Race for CRF

By the spring of 1981, the race to identify the chemical structure of CRF had reached a fever pitch. For twenty-six years, the "elusive" factor had been the subject of countless failed attempts by laboratories across the globe.[3] On April 1, 1981, the psychological strain within the Vale laboratory manifested in a moment of elaborate deception. Ron Kaiser, a technician for Jean Rivier, recognized the unbearable tension and decided to leverage the date to provide a moment of dark humor.[3]

Kaiser carefully doctored a table of contents from a forthcoming issue of *Science*. He inserted a fake paper titled "Discovery of Corticoliberin" (an alternative name for CRF) and attributed the discovery to Andrew Schally's rival team.[3] When the document was presented to the team, the initial reaction was one of crushing defeat. Vale, however, possessed a keen awareness of the competitive landscape and the calendar. He quickly identified the hoax, but the momentary shock served to illustrate exactly how much was at stake.[3] The scene underscores the reality of "scooping" in high-level science; being second meant total erasure from the primary discovery record. Shortly thereafter, Vale would indeed use the Beckman spinning cup sequencer to definitively solve the 41-amino acid sequence of ovine CRF, finally ending the quarter-century search.[3]

SCENE #3: The Ectopic Tumor Pivot to Growth Hormone Releasing

Factor (GHRF)

In 1982, Vale demonstrated his "translational" intuition by solving the mystery of Growth Hormone Releasing Factor (GHRF) not through the brain, but through a rare clinical manifestation. While GHRF had been hypothesized to exist in the hypothalamus, its concentration there was too low for traditional sequencing methods of the time.[7] The breakthrough occurred when Vale collaborated with clinical endocrinologists Michael Thorner and Lawrence Frohman, who were treating patients with acromegaly caused by ectopic tumors.[7]

Discovery Step	Detail	Source
The Source	Ectopic tumor from a patient with acromegaly	7
The Rationale	Tumors produced higher concentrations of GHRF than the brain	7
Sequence Length	44-amino acid peptide	7
Collaborators	Michael Thorner (UVA), Lawrence Frohman (UIC)	8
Technique	HPLC and Microsequencing	4

The identification of GHRH from a pancreatic islet tumor was a pivotal moment. It proved that a hypothalamic hormone could be produced "out of place" by a tumor, leading to the same physiological effects as the brain-derived version.[7] This discovery completed the characterization of the primary hypothalamic-pituitary regulators and allowed for the synthesis of GHRH for clinical use in treating growth disorders.[2]

SCENE #4: The Declaration of Independence and the Founding of the PBL (1978)

The transition of Wylie Vale from Roger Guillemin's star protégé to an independent scientific force occurred in 1978. Following Guillemin's 1977 Nobel Prize, the administrative and intellectual landscape at the Salk Institute shifted.[2] Vale, alongside Jean and Catherine Rivier and Marvin Brown, made the decisive move to establish the Peptide Biology Laboratory

(PBL).[2]

This was not merely a change in title; it was a physical and strategic separation. The newly formed PBL was relegated to temporary wooden barracks located west of the main Louis Kahn-designed Salk building.[3] While the space was modest and utilitarian, it offered Vale the autonomy to focus exclusively on the "holy grail" of CRF, which had eluded Guillemin for decades.[2] The "temporary" nature of the labs did not hinder their output; the barracks became the birthplace of the molecular stress response field.[3] This scene marks the birth of Vale as an institutional leader who would eventually head the Clayton Foundation Laboratories for Peptide Biology for thirty-four years.[12]

SCENE #5: The Intensely Competitive Morning Rituals

Before the scientific day began in the lab, the culture of the Vale group was forged on the racquetball and tennis courts of La Jolla. George Koob, a long-time collaborator, describes the atmosphere at the Salk as an "intensely competitive scientific playground".[3] Vale was a strong-willed athlete who brought the same "chess master" intensity to sports as he did to peptide sequencing.[3]

Lab Culture Element	Description	Source
Morning Activity	Racquetball or tennis	3
Atmosphere	"Intensely passionate and energetic"	3
Interpersonal Dynamics	High competition followed by high collaboration	3
Attire/Gear	Surfboards and scuba gear often seen in the lab barracks	2

This scene provides a window into the "youthful exuberance" that Vale maintained throughout his career.[2] The physical competition served as a release for the intellectual pressure of the "Nobel Duel" and the grueling extraction work. It also reflected Vale's philosophy that successful science required both "grit and zest".[3]

B) Direct Quotes From the Scientist

Quote #1: The Vision of Future Medicine

> "I work in the Neuroendocrinology Laboratory at the Salk Institute in a field that I think is opening up another new era of medicine, one in which emphasis is not on curing infections but on putting right some of the body's own mistakes. Our laboratory is working with small molecules that are made in our brains and which are used to regulate a host of activities in the body." [2]

Context: This statement was recorded during the early phase of Vale's career, likely in the early 1970s. It reveals a forward-thinking "translational" perspective, anticipating the shift from infectious disease medicine to the management of chronic physiological and psychological dysfunctions.[2]

Quote #2: The Standard of Performance

> "Effort only counts through the 3rd grade, Catherine. After that, only results matter." [3]

Context: Spoken to his collaborator and mentee Catherine Rivier during a period of scientific stagnation. It encapsulates Vale's uncompromising commitment to definitive characterization over mere attempt.[3] It was a "Vale-ism" used to motivate and focus his team during the long years of searching for CRF.[3]

Quote #3: The Strategic Philosophy

> "I skate to where the puck is going to be, not where it has been." [12]

Context: While famously a Wayne Gretzky quote, Vale adopted this as his professional mantra. It informed his transition from classical physiology to molecular neurobiology and his early involvement in the biotechnology industry (Neurocrine Biosciences, Acceleron Pharma).[12]

Quote #4: The Experience of Self-Experimentation

> "It felt like you were going to die—you wet your pants, your heart raced, you become very light-sensitive!" [3]

Context: Vale's description of being injected with thyrotropin-releasing factor (TRF) under medical supervision during the early trials at Baylor. This was a common, though high-risk, practice during the "cowboy" era of endocrinology to determine the physiological effects of newly synthesized peptides.[3]

Quote #5: On the Translational Goal

> “Our laboratory is working with small molecules that are made in our brains and which are used to regulate a host of activities in the body.” [2]

Context: This recurring theme in Vale’s lectures emphasized that peptides were not just "brain chemicals" but systemic regulators that could be harnessed for drug development.[2]

C) Human Reality Details

The persona of Wylie Vale was shaped by a combination of his Texan upbringing and the elite scientific environment of the Salk Institute. The following details provide a sourced account of his human reality:

1. **Architectural Heritage**: Vale was the son of Wylie Walker Vale Sr., a prominent Houston architect. The elder Vale was known for his "Contemporary Country" style, designing over 400 residential and institutional buildings.[14] This background in structural design may have contributed to Wylie Jr.’s "chess master" approach to molecular structures.[3]
2. **Long-Term Partnership**: He met his wife, Betty, while in high school. They were married for over 40 years, raising two daughters, Elizabeth and Susannah.[1]
3. **Texan Spirit**: Despite spending the majority of his career in Southern California, Vale remained a "true Texan in spirit," often wearing cowboy boots and teaching international trainees "the local vernacular" and taking them to rodeos.[2]
4. **"Kefi" and Joy**: Colleagues frequently used the Greek term *kefi* to describe his spirit of joy, passion, and love for life.[3] He was a "lover of wine" and "ready for adventure".[12]
5. **Mentorship as Customization**: Vale was noted for his "social intelligence," allowing him to tailor his mentoring to the individual needs of his staff. He encouraged "provocative thought" and gave researchers the "license and wherewithal to pursue it".[2]
6. **The Racquetball Court**: His mornings were defined by "intensely competitive" racquetball sessions, a habit he maintained for over 40 years at the Salk.[3]
7. **The Smell of Discovery**: During the isolation of TRH and CRF, Vale and his team worked with such high volumes of biological extracts and organic solvents that they "smelled so bad that people in the supermarket would run away".[3]
8. **Fly-Fishing Passion**: In the final decade of his life, he became an ardent fly-fisherman, a hobby that allowed him to connect with "raw nature" in places like Hawaii.[12]
9. **Disarming Humor**: He possessed a "biting wit" and an ability to "intertwine scientific discourse with the recounting of day-to-day humorous life events," which often left colleagues "completely disarmed".[16]
10. **The "Third Grade" Rule**: He held a firm belief that in professional science, effort was secondary to results, a philosophy he instilled in his students to prepare them for the rigors of the field.[3]
11. **Technological Curiosity**: Weeks before his death, he was enthusiastically showing friends pictures of his granddaughter on his iPhone, reflecting his lifelong habit of being an early adopter of new tools.[12]

12. **Final Resting Place**: Vale died unexpectedly in his sleep while on vacation in Hana, Maui, in a house located at the edge of a rainforest that he and Betty were remodeling.[1]
13. **Global Network**: He was a "longstanding friend of the Hellenic Endocrine Society" and felt "at home in Greece," where he was a frequent and celebrated lecturer.[16]

D) Science Connection (Intro Biochemistry: Signal Transduction and Peptide Hormones)

The work of Wylie Vale provides a concrete bridge for students of BIOL 3030 into the mechanisms of **Signal Transduction** and the **Biochemistry of Peptide Hormones**.

Precise Contribution

Wylie Vale isolated, sequenced, and characterized the primary hypothalamic peptides—Corticotropin-Releasing Factor (CRF) and Growth Hormone Releasing Factor (GHRF)—and their respective G-protein coupled receptors (GPCRs), thereby elucidating the molecular initiation of the mammalian stress and growth responses.[1]

Teachable Bridge Explanation

In introductory biochemistry, students study how cells respond to external stimuli via **ligand-receptor interactions**. Vale's discovery of CRF (a 41-amino acid peptide) and its receptors (CRFR1 and CRFR2) illustrates this perfectly.[6] These hormones are synthesized as large **prohormones** and processed through the secretory pathway. When a stressor is perceived, CRF is released from the hypothalamus and travels to the anterior pituitary, where it acts as a ligand for a **G-protein coupled receptor (GPCR)**.[6]

The binding of CRF to its receptor triggers a conformational change that activates a **stimulatory G-protein (G_s)**, which in turn activates **adenylyl cyclase**. This enzyme converts ATP into **cyclic AMP (cAMP)**, a second messenger that initiates a phosphorylation cascade. This cascade ultimately leads to the release of **Adrenocorticotropic Hormone (ACTH)** into the general circulation, which then targets the adrenal cortex to produce cortisol.[1] This is the fundamental "bridge" from a neural signal to a systemic hormonal response.

Scientist's Framing/Analogy

Vale and his colleagues referred to this system as the "central switchboard" of the body, where the hypothalamus acts as the operator, receiving nerve signals from the brain and "plugging in" the appropriate hormonal response to regulate metabolism, growth, and stress.[1]

E) Complications, Disputes, or Controversies

Item 1: The "Nobel Duel" Rivalry

What's agreed upon: Roger Guillemin and Andrew Schally were engaged in a decades-long, intensely competitive race to isolate hypothalamic hormones, for which they shared the 1977 Nobel Prize.5 Wylie Vale was a primary collaborator of Guillemin during the discovery of TRH, GnRH, and somatostatin.2
What's disputed: The level of personal animosity vs. professional competition remains a subject of debate. Nicholas Wade's book The Nobel Duel portrays the rivalry as "intensely bitter".5 However, later memoirs from Guillemin regarding Vale suggest a relationship that, while competitive, maintained high levels of mutual admiration and shared personal interests like wine.2
Sources:.2

Item 2: The CRF vs. Vasopressin Identity Crisis

What's agreed upon: Since the 1950s, scientists knew a "factor" in the hypothalamus stimulated the pituitary, but its identity was unknown. Many argued that Arginine Vasopressin (AVP) was the actual CRF because AVP also stimulates ACTH release.3
What's disputed: Vale had to prove that his 41-amino acid peptide was the primary physiological regulator. This required developing a specialized pituitary bioassay to distinguish between the effects of AVP and the newly isolated CRF.3 The dispute was only fully resolved when Vale's group produced a CRF antibody that could block the stress response in vivo, proving that CRF, not vasopressin, was the essential driver.3
Sources:.3

Item 3: The "Sequel" to the Nobel Duel (Activin)

What's agreed upon: In 1986, both Vale's group and Guillemin's group (now operating independently) published papers in the same issue of Nature describing the discovery of Activin.2
What's disputed: This simultaneous publication is often described by historians as the "sequel" to the original Nobel Duel, as the former mentor and protégé were now direct rivals for priority in the discovery of new peptide families.19
Sources:.19

F) "Not on Wikipedia" Gold

1. **The TRF Self-Injection**: Before modern Institutional Review Boards (IRBs), Vale was an active participant in his own research. He was injected with TRF and experienced a dramatic "near-death" reaction including racing heart and involuntary physiological responses, a detail rarely included in sanitized professional biographies.[3]
2. **Architectural Influence on Peptide Science**: Vale's father helped organize Piney Point Village in Houston and was a pioneer of the "Contemporary Country" style.[14] Colleagues noted that Wylie Jr. approached scientific problems "like a chess master" and with an "architectural" sense of how molecular components fit together.[3]
3. **The Grocery Store Pariahs**: The extraction of hormones from half a million sheep

required so much mercaptoethanol and organic solvent that the team's clothing and skin absorbed the scent. This led to them being physically avoided by the public in La Jolla supermarkets during the early 1970s.[3]

4. **"Effort Only Counts Through the 3rd Grade"**: This specific mantra was Vale's way of enforcing a high-performance culture. He used it to remind researchers that "trying" was insufficient without "finding".[3]
5. **Greek Cultural Immersion**: Vale was more than just a visitor to Greece; he was an honorary member of the Hellenic Endocrine Society and was known for giving lectures that mixed high-level peptide biochemistry with humorous anecdotes of Greek daily life, disarming even his toughest critics.[16]

G) Data Tables

Table 3: Summary of Key Molecular Characterizations by Wylie Vale

Molecule	Year	Amino Acid Count	Primary Target	Clinical Significance
CRF	1981	41	Pituitary (Corticotrophs)	Anxiety, Depression, IBS
GHRF	1982	44	Pituitary (Somatotrophs)	Growth disorders, Acromegaly
Inhibin	1985	Heterodimer	Pituitary (Gonadotrophs)	Reproductive diagnostics
Activin	1986	Homodimer	Multiple tissues	Cell differentiation, Cancer
Urocortin 2	2004	38	Heart/Vasculature	Congestive heart failure

Table 4: Professional Milestones and Institutional Leadership

Period	Role/Position	Institution	Key Activity
1964-1968	Ph.D. Student	Baylor College of Medicine	TRH extraction research
1970-1978	Associate Professor	Salk Institute	Collaboration with Guillemin
1978-2012	Lab Head	PBL (Salk Institute)	Discovery of CRF and GHRF
1992	Co-Founder	Neurocrine Biosciences	Translational drug development
2003	Endowed Chair	Helen McLoraine Chair	Molecular Neurobiology research

H) Flagged “Myth-Risk” Details

Claim: The discovery of GHRH was a purely "accidental" find from a single tumor.

- **Status**: High Risk of Over-simplification.
- **Nuance**: While the tumor was the *source*, the discovery was part of a decades-long strategic search. Vale had already established the bioassays and the theoretical framework; the tumor provided the "concentrated" material that the brain could not.[4]

Claim: Vale and Guillemin were in a "hostile" feud after 1978.

- **Status**: Medium Risk.
- **Nuance**: Competition for credit was fierce (especially over Activin), but private correspondence and memorials show a continued bond over shared aesthetics and passions (e.g., wine, Pinot Noir).[2]

LENS RECOMMENDATION

Lens scoring rubric (0–3 each)

Lens	Score	Justification

Eureka	3	The 1981 isolation of CRF is the definitive "Eureka" of the field, solving a 26-year-old mystery with a specific technological breakthrough (the sequencer).[3]
Failure	1	While the "long search" involved setbacks, there is no single well-documented "catastrophic failure" that pivoted his life.
Mentor	3	The arc from Guillemin's student to his "sequel" rival is a perfectly documented transition of power and independence.[2]
Controversy	2	The rivalry with Schally and the split from Guillemin provide tension, but were often handled with professional decorum rather than public scandal.[5]
Human Detail	3	The combination of Texan "cowboy" spirit, the "smelly" lab era, and the architectural father provides deep, non-standard texture.[3]
Last Experiment	2	His death in Hawaii while remodeling a home is a poignant closing image, but lacks a specific "final scientific act" scene.[1]

Primary recommended lens: Eureka (Discovery Crystallized)

- **Why it fits**: The isolation of CRF in 1981 serves as the backbone of his narrative. It

connects the "smelly" extraction years of the 70s to the molecular era of the 80s.[3]
- **Best opening scene**: Scene #1 (The extraction marathon/smell detail) provides the most vivid sensory cold open.
- **One quote to use early**: Quote #2 ("Effort only counts through the 3rd grade").
- **Cautions**: Avoid making the Schally rivalry the *only* story; focus on the technical perseverance within the Vale lab.

Backup lens: Mentor (The protégé's independence)

- **Why it fits**: It frames the 1978 lab split as the pivotal moment where Vale "stepped out" to solve the one problem his Nobel-winning mentor couldn't crack.[2]

Detailed Analytical Narrative: The Molecular Architecture of Stress

The scientific life of Wylie Vale is a study in the transition from the macroscopic to the microscopic. His father, an architect of physical spaces, taught him that structure dictates function.[15] In the biological realm, Vale applied this to the peptide. The "corticotropin-releasing factor" (CRF) had been a ghost in the machinery of the brain since Harris first hypothesized it in 1948.[4] For decades, researchers could see its *effects*—they could inject a brain extract into an animal and see the pituitary respond—but they could not see the *molecule* itself.

The Brute Force Era

The 1960s and early 70s were characterized by what might be called "brute force biochemistry." To find a few micrograms of a hypothalamic hormone, one needed tons of brain tissue. Vale’s early career at Baylor was defined by this logistical nightmare. The sheer scale of the sheep hypothalamus collection—over 500,000 units—required a "factory" mindset.[3] The organic solvents used in the purification, such as petroleum ether and triethylammonium phosphate, were not just laboratory tools; they were environmental constants that permeated the very skin of the researchers.[3] This era taught Vale that scientific breakthroughs were as much about physical endurance as intellectual insight.

The Shift to Molecular Precision

The move to the Salk Institute in 1970 and the subsequent founding of the Peptide Biology Laboratory (PBL) in 1978 marked Vale's shift toward the "Gretzky" philosophy of anticipating where the field was going.[2] By 1981, technology had finally caught up to his ambition. The Beckman spinning cup sequencer allowed him to crack the 41-amino acid code of CRF.[3]

Mathematically, the diversity of a 41-amino acid peptide is staggering. If each position can be occupied by any of the 20 standard amino acids, the number of possible sequences is

20^{41}. Vale’s task was to find the one sequence that existed in the hypothalamus. The successful characterization of this sequence:

$$S-E-E-P-P-I-S-L-D-L-T-F-H-L-L-R-E-V-L-E-M-A-R-A-E-Q-L-A-Q-Q-A-H-S-N-R-K-L-M-E-I-I$$

was the "Eureka" moment that allowed for the synthesis of the hormone and the creation of antibodies.3

The Translational Turn

Vale was not content with purely academic discovery. He understood that the "central switchboard" of the brain was a prime target for pharmacological intervention. His founding of Neurocrine Biosciences in 1992 was a manifestation of his belief that these peptides were the key to treating "the body's own mistakes".[2] By identifying the specific G-protein coupled receptors (CRFR1 and CRFR2), he provided the pharmaceutical industry with the precise "locks" for which they could design synthetic "keys".[6] His later work on urocortins and their role in cardiac function showed that the stress response was not just a brain event, but a systemic one, influencing everything from heart muscle contraction to appetite.[1]

Legacy and Human Continuity

At his death in 2012, Vale left behind a body of work that included over 600 peer-reviewed papers and the discovery of more than a dozen novel hormones.[1] But his human legacy was found in the "youthful curiosity" he maintained and the "social intelligence" he used to guide over a hundred students and fellows.[2] He lived as he worked—at the edge of the rainforest, remodeling the structures around him, always looking for the next breakthrough, always skating to where the puck was going to be.[12]

Works cited

1. Wylie Vale, Salk scientist, pioneer and leader, dies at 70, accessed January 21, 2026, https://www.salk.edu/news-release/wylie-vale-salk-scientist-pioneer-and-leader-dies-at-70/
2. In Memoriam: Wylie Vale, Jr., Ph.D. (1941–2012) - PMC - NIH, accessed January 21, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC5416993/
3. Minireview: CRF and Wylie Vale: A Story of 41 Amino Acids and a ..., accessed January 21, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC3359597/
4. Key Role of CRF in the Skin Stress Response System - PMC - PubMed Central, accessed January 21, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC3857130/
5. The Nobel Duel - Asimov Press, accessed January 21, 2026, https://press.asimov.com/articles/nobel-duel
6. CORTICOTROPIN-RELEASING FACTOR: PHYSIOLOGY, PHARMACOLOGY, AND

ROLE IN CENTRAL NERVOUS SYSTEM DISORDERS - ACNP, accessed January 21, 2026, https://acnp.org/wp-content/uploads/2017/11/CH7_91-108.pdf
7. Handbook of Depression - ResearchGate, accessed January 21, 2026, https://www.researchgate.net/profile/Gerrit-Glas/publication/28647791_A_Conceptual_History_of_Anxiety_and_Depression/links/0deec52a6358706b6e000000/A-Conceptual-History-of-Anxiety-and-Depression.pdf
8. Regulation of GH and GH Signaling by Nutrients - MDPI, accessed January 21, 2026, https://www.mdpi.com/2073-4409/10/6/1376
9. Standards for undergraduate medical education - Society for Endocrinology, accessed January 21, 2026, https://www.endocrinology.org/media/1290/103.pdf
10. Chappelle: - Endocrine Society, accessed January 21, 2026, https://www.endocrine.org/-/media/endocrine/files/community/sawin/lawrencefrohman_transcript.doc
11. Wylie Vale - Wikipedia, accessed January 21, 2026, https://en.wikipedia.org/wiki/Wylie_Vale
12. Wylie W Vale - PMC, accessed January 21, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC3499721/
13. Obituary Wylie W. Vale, Ph.D. - ACNP, accessed January 21, 2026, https://acnp.org/wp-content/uploads/2017/10/ValeObituary.pdf
14. August Mod of the Month #2 - HoustonMod, accessed January 21, 2026, https://www.houstonmod.org/august-mod-of-the-month-2/
15. Protected Landmark Designation Report - CITY OF HOUSTON, accessed January 21, 2026, https://www.houstontx.gov/planning/HistoricPres/landmarks/08PL64_Bouknight_House_3023_Ella_Lee.pdf
16. Wylie Walker Vale, Jr. - Hormones.gr, accessed January 21, 2026, http://www.hormones.gr/748/article/wylie-walker-vale-jr.html
17. A novel corticotropin-releasing factor receptor splice variant exhibits dominant negative activity: a putative link to stress-induced heart disease - PubMed Central, accessed January 21, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC2704597/
18. REMEMBERINGWylie 03 - Inside Salk, accessed January 21, 2026, https://inside.salk.edu/wp-content/uploads/2016/05/inside_salk_03-12.pdf
19. In celebration of 100 exciting years of the Japan Endocrine Society - PMC - NIH, accessed January 21, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11867630/
20. Dr. Wylie W. Vale, Jr. Memorial Lecture celebrates and inspires research, accessed January 21, 2026, https://blogs.bcm.edu/2025/07/10/from-the-labs-dr-wylie-w-vale-jr-memorial-lecture-celebrates-and-inspires-research/